

Comparing SDLC Approaches

Step 1: Analyze the project scenario and identify its key characteristics

Feature	Best Fit(s) (Waterfall, Agile, SAP Activate) and Brief Rationale
Approach	<i>Agile's flexibility suits the evolving AI, while SAP Activate might be used for integration with existing systems. Waterfall is too rigid.</i>
Requirements	<i>Agile. High-level integration goals are defined upfront, but detailed user stories ,AI personalization, accessibility features, are refined iteratively based on continuous feedback.</i>
Planning	<i>Agile/hybrid. Continuous planning through sprint backlogs aligned with the update schedules and maintenance windows of the existing LMS.</i>
Development	<i>Agile, Incremental, sprint-based development allows building cloud-based AI components and API connectors step-by-step without disrupting current platform functionality.</i>
Testing	<i>Agile hybrid. Continuous integration, regression testing for the legacy system, and WCAG accessibility testing with PWD focus groups. Synthetic employee data must be used to test AI without violating privacy regulations.</i>
Customer Involvement	<i>Agile. Active engagement with HR, IT administrators, and disabled employee focus groups during sprint demos ensures early feedback on usability and accessibility.</i>
Change Management	<i>Agile/SAP activate. Phased rollout starting with a Minimum Viable Product to a pilot group, allowing smooth user adoption, continuous feedback, and tailored training.</i>
Risk Management	<i>Agile. High-risk factors, such as cloud AI latency, legacy API bottlenecks, data privacy compliance, and accessibility blockers, are identified and mitigated early in short iterations.</i>
Documentation	<i>Hybrid. Living, lightweight documentation for sprint user stories combined with formal API and integration docs required to maintain the existing platform long-term.</i>
Suitable Projects	<i>Agile/Hybrid. Ideal for brownfield expansion projects that integrate innovative cloud/AI technology into established enterprise systems.</i>
Project Speed	<i>Agile. Delivers rapid value through an MVP, enabling quick user feedback on language learning while maintaining system stability through short release cycles.</i>
Focus	<i>Agile. Prioritizes continuous user value, accessibility compliance, data privacy, and system seamlessness over rigid adherence to an initial fixed plan.</i>

Step 2: Compare the three SDLC Approaches

Step 2.1: Traditional (Waterfall)

Feature	Considerations
Strengths for HablaMundo	<i>Upfront architectural planning, comprehensive documentation, predictable milestones</i>
Weaknesses for HablaMundo	<i>High rigidity, late testing phase, delayed user feedback</i>

Step 2.2: Agile

Feature	Considerations
Strengths for HablaMundo	<i>Iterative accessibility testing, flexibility for AI features, early value delivery</i>
Weaknesses for HablaMundo	<i>Risk of scope creep, architectural friction, high resource demand</i>

Step 2.3: SAP Activate

Feature	Considerations
Strengths for HablaMundo	<i>Build for systems extensions, hybrid flexibility, integrated quality and migration controls</i>
Weaknesses for HablaMundo	<i>Administrative overhead, learning curve, rigid core assumptions</i>

Step 3: Choose the most appropriate approach for the scenario and justify the selection

The most appropriate methodology for the HablaMundo project is a Hybrid Agile approach incorporating structured extension governance from frameworks like SAP Activate. While pure Waterfall is too rigid to accommodate evolving cloud-AI algorithms and delays critical accessibility testing, pure Agile risks bypassing legacy LMS architectural constraints. This hybrid strategy balances flexibility and control by utilizing short Agile sprints so disabled employee focus groups can continuously test accessibility features, while applying SAP Activate's Fit-to-Standard governance to protect existing LMS architecture, data security, and API integration standards. Additionally, iterative development supports the ongoing refinement of cloud-based AI tools, and deploying a Minimum Viable Product to a pilot user group validates employee data mapping early, de-risking the global rollout.